



metang

a gated evolutionary RHEA agent

Manan Gupta · The Harker School, San Jose

Planet Wars RTS · IEEE Conference on Games 2026 · mnn@yogins.com

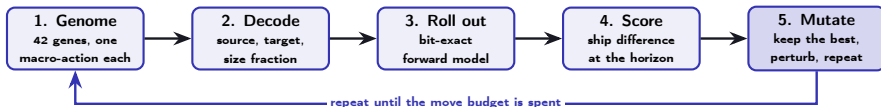
12 entries on the spring-2026 league, all from one lineage

#	agent	μ	#	agent	μ
2	metagross	45.555	12	masked-royal	38.230
3	metang	45.429	13	incineroar-z	37.064
5	beldum	43.193	17	incineroar	33.550
7	onix	39.762	21	torracat	29.737
9	mega-heracross	39.449	32	litten	23.953
10	wolfes-incineroar	39.388	61	jasmines-metagross	<i>unrated</i>

Official league board, 2026-09-17 19:14 UTC, 61 ranked entries. TrueSkill; ranks are by $\mu - 3\sigma$, values shown are μ . The league is live and ranks move.

The agent: rolling-horizon evolutionary search

The game. Two players hold planets that grow ships every tick. A move sends part of one planet's garrison to another; arriving ships reinforce a friendly planet or capture a weaker one. A player with no planets left loses; otherwise the game runs to 2,000 ticks. Real time, 50 ms per decision, randomly sampled maps.



beyond the 42 searched ticks, a fixed 160-tick heuristic tail finishes the rollout

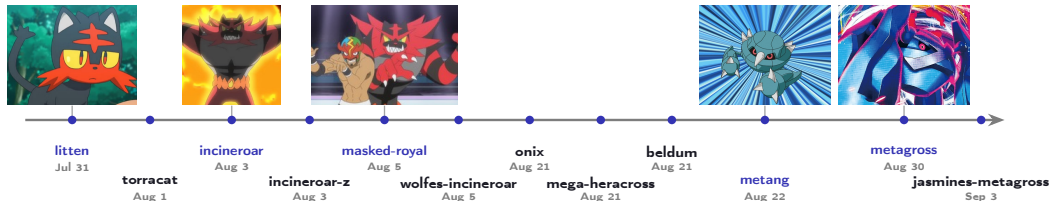


Shift buffer: execute the first gene, slide the plan left, append one fresh gene. The surviving plan is the memory.

Champion configuration, verbatim:

```
rhea:horizon=42,reinforce=true,opp=None,warmup=true,firstMove=0.6,sizeFloor=true,  
discount=0.93,squash=1000,projectDef=true,beldum=true,guardReturn=true,fixed=1200
```

One lineage: twelve entries in five weeks



All twelve entries in submission order. Each shipped generation takes the next name in its evolution line.

Jul 31 to Aug 5: hand-gated RHEA

litten and five gated successors: a rolling-horizon evolutionary agent, each change accepted only after a pre-registered head-to-head against its predecessor.

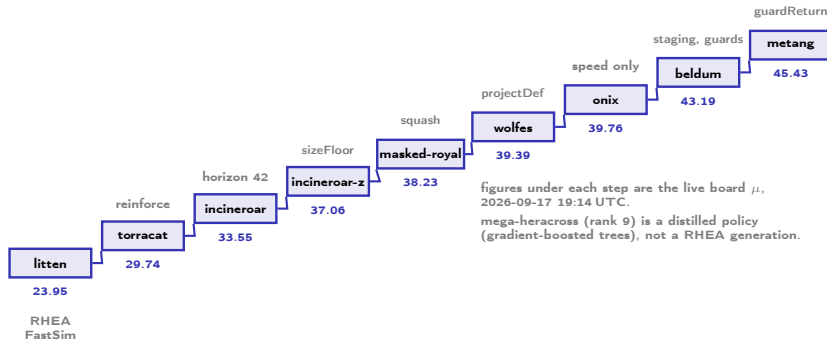
Aug 21 to 22: the factory behind the gate

An LLM mutation loop produced beldum and then metang, the converged local optimum. Alongside it, an engine-speed build (onix) and a distilled policy (mega-heracross, gradient-boosted trees).

Aug 30 to Sep 3: budget variants

metagross and jasmynes-metagross are metang with container environment lines changed and no code changes. Those two are the second half of this deck.

Eight gated wins on one baseline, one pattern

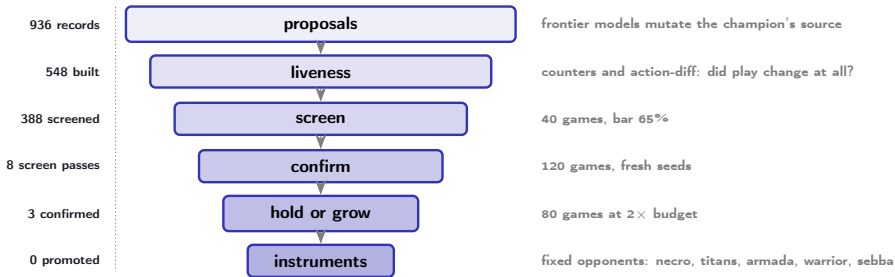


The pattern

Seven of the eight gated wins changed **how the board is priced**: which planets are targeted, how a target's defence at arrival is computed, or what an existing gene decodes to in a dead or under-sized case. The eighth, onix, changed only the simulator's speed. None changed the search operator, and none added an action to the vocabulary.

Method: a (1+1) loop behind a pre-registered gate

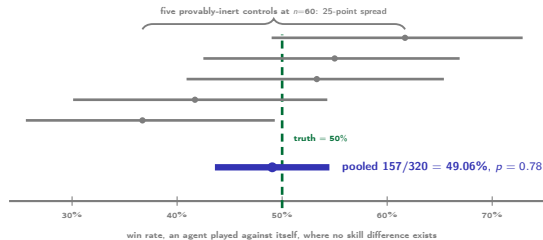
final audited run, 13,780 games



Over the campaign a handful of candidates cleared this gate, and each became a shipped generation. The rail on the left is the final audited factory run: 936 records in, 0 promotions out. Every threshold was written down before the confirming data existed.

Prior art: the WCCI 2026 winner (the `anti` entries, ranks 8, 11, 14 and 16 on this board) published an LLM-mutation loop for this competition first. Our contribution is the acceptance gate, not the loop.

Why the gate: the objective is very noisy



Signal 1: model confidence is empty

The proposing model's stated confidence does not predict whether the idea wins games.

$$r = 0.064 \quad (\rho = 0.041, n = 388)$$

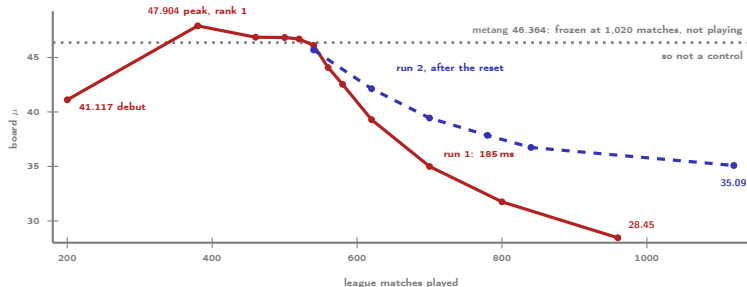
Signal 2: the screen's own ordering inverts

candidate	screen 40g	confirm, vs champion
tailscan-66	67.5%	+3.5 pp
sol staging	70.0%	+0.2 pp
c661 arrival	77.5%	-0.6 pp

120-game confirms vs the champion's own 63.1%; tailscan pooled over two.

Under that floor a 40-game screen at a 65% bar promotes noise about 4% of the time. Generate with the model; decide in the arena.

A negative result: more move budget did not help, twice



Two declines, and a comparison that looked cleaner than it was

metagross is metang with two container lines changed: league budget 150 ms to 185 ms, qualifier 35 ms to 30 ms. The local evidence was never significant: doubling rollouts scored 52.7% [47.0, 58.2] over 300 games, an interval that contains 50. On the league it lost rating at every rebuild after its peak, down to 28.45. The league was **reset** on 2026-09-02; metagross re-seeded at 45.69 and declined again, to 35.09 by 1,120 matches. **Two declines, same code, same variable, across an independent reset.**

metang could not serve as a control here. Its match count was frozen at 1020 from 2026-09-01, so it was not playing under the same conditions as metagross. The apparent comparison was really one active agent against a stopped clock. The cause turned out not to be the budget size at all. The next slide identifies it.

The cause, found: behaviour that depended on container age

The organiser supplied the missing variable: the round-robin script reset containers *less often* than the regular league. Our agent chose its move budget from two signals, and one was **process-global**: the first 12 games a container ever played used a safe 30 ms budget, every game after used 185 ms. Frequent resets kept re-entering the safe window. Infrequent resets passed 12 permanently, so every later game ran at full budget, missed the per-move deadline, and our moves became doNothing.

	metagross (ramp 12, 185 ms)		jasmynes-metagross (no ramp, 45 ms)	
games	wins	deadline misses	wins	deadline misses
1–8	8/8	2	8/8	7
9–16	4/8	7,226	8/8	15
17–24	0/8	15,469	8/8	5
total	12/24	22,697	24/24	27

Official runner, 50 ms deadline, one container per arm, greedy heuristic opponent. A deadline miss is one late move; a game has hundreds.

What shipped, and what did not

The fix sets the ramp length to zero, so the counter is never consulted and the budget depends only on `maxTicks`, a property of the game rather than of the container. The league budget drops to 45 ms. That reduction costs nothing measurable: 45 ms versus 150 ms scored 50.0% [37.7, 62.3] over 60 mirrored games.

Parallel search was tested and **rejected**. `threads=2` passed a 40-game screen at 70.0% [54.6, 81.9] and then failed its 120-game confirm on a fresh seed at 55.0% [46.1, 63.6]. The screen was the same mirage the gate exists to catch. **jasmynes-metagross is the bug fix alone: three environment lines, zero code changes.**

Where things stand

#	agent	μ	matches
1	tail-small	49.602	2,177
2	metagross	45.555	2,009
3	metang	45.429	2,647
4	tail-warm-start	44.660	420
5	beldum	43.193	980
61	jasmines-metagross	<i>unrated</i>	0

Board as of 2026-09-17 19:14 UTC. TrueSkill; ranks by $\mu - 3\sigma$, values shown are μ . Our entries in bold. The league is live and ranks move.

The league recovered. Our fix has not been tested by it.

The league has since resumed ordinary operation, and both legacy entries recovered: metagross and metang sit at ranks 2 and 3, separated by 0.13 and 4.05 behind the leader. That recovery is **consistent with** the diagnosis on the previous slide, since frequent container resets keep the counter inside its safe window. We have not confirmed what changed operationally, and we make no stronger claim than that.

jasmines-metagross passed its qualifier cleanly (100%, zero deadline misses) and is accepted, but has not yet been scheduled by the league: zero matches to date. It carries no rating, good or bad, until it plays.

Representative entry requested: **jasmines-metagross (submission #195)**. metagross and metang are the same search code at other move budgets. Code, paper and these slides: github.com/mnn31/planet-wars-agent (private; the organiser has read access).